DBMS Module 4

- Introduction to NoSQL Concepts

Stands for Not only SQL.

Non tabular database systems that stores data differently. They are built to manage large amount of unstructured and semi-structured data with flexible schema and high scalability.

egs: MongoDB, Neo4j, Cassandra.

- Types of NoSQL databases

1. Document based:

Stores data as documents (JSON, BJSON, XML) with each having different field flexibility. It is used in websites, blogs etc.

Closely matches application data objects, supports indexing for faster access.

```
{ "_id": "abc",
  "Name": "Anna"
}
```

Features: flexible schema, faster creation & maintenance, no foreign keys

eg: MongoDB, Document DB

2. Key Value Pair:

Hash table that stores a pointer (key) which then points to value (JSON, lists, BLOB) that stores some form of information. Used to handle heavy load. Doesn't support structured schemas, relationships. Values can't be queried, only key can be searched.

Key	Value
Anna	12
Anit	14
Aby	IT

eg: Redis, Dynamo

3. Column-Based:

Stores data in columns instead of rows. It helps in reading large volumes of data quickly and efficiently. Aggregates values of given column. They use the concept of keyspace which contains all column families. Row key is used as identifier.

eg: Casandra, Cosmo DB

Id	Name	Course
1	Alice	A
2	Ben	B
3	Chris	C

Id	Name
1	Alice
2	Ben
3	Chris

Id	Course
1	A
2	B
3	C

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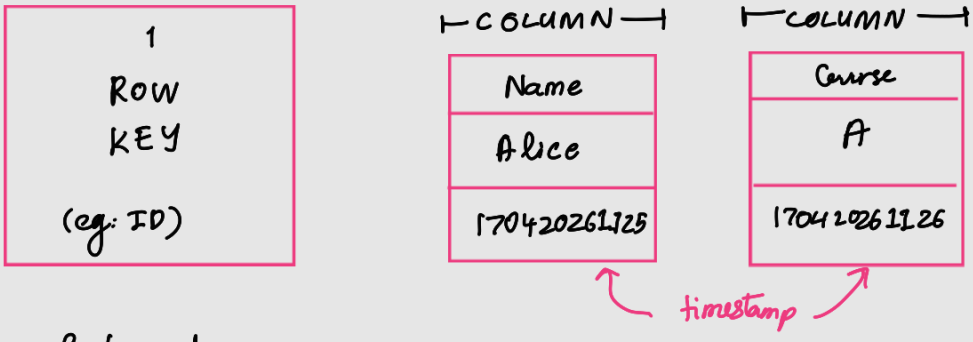
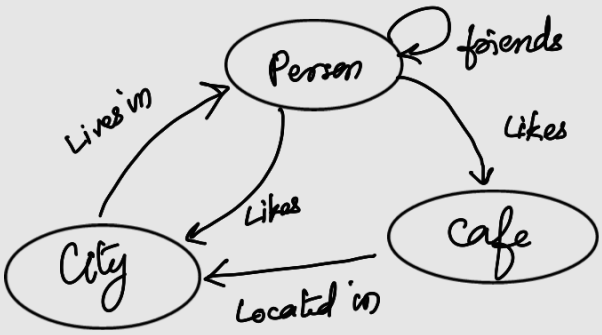


4. Graph based:

Stores data as nodes (data) and edges (relationship). Best for understanding relationships b/w data. Used in recommendation engines, social networks.



eg: Neo4j, ArangoDB

- CAP Theorem / Brewer's theorem:

  C: Consistency

  A: Availability

  P: Partition tolerance

Consistency: Every read receives the most recent write or an error. All nodes in the system have the same data at any given time. It ensures all clients have single, upto date view of data.

- All nodes see same data at same time, irrespective of which node processes request.
- If write done, all subsequent reads have updated value

eg: Bank transactions.

Availability: Every request receives a response, without guarantee that it contains most recent writes. System remains operational even during failures.

- Ensures every req receives a response.

eg: Website (some pages undergoing maintenance)

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Partition Tolerance: System continues to operate despite n/w partitions that may cause messages to be delayed / lost b/w nodes. It ensures that system can withstand failures of n/w components without entire system failing

- System continues to operate despite of failure of part of system
- Can tolerate n/w failures.

eg: Global distributed databases → region communication issues

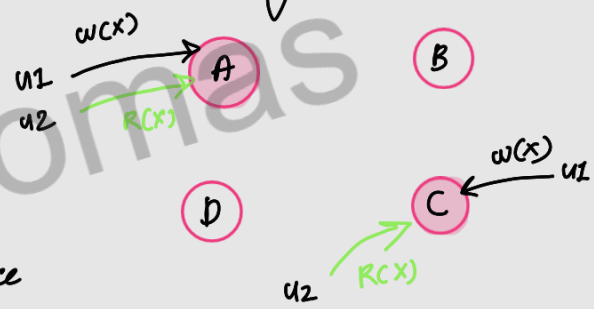
According to CAP theorem, a distributed system can only prioritize two out of these three guarantees simultaneously.

CA system, CP system, AP system.

No SQL databases often adhere to CAP theorem based on their design goals

- BASE theorem

Basically Available Soft-state Eventual consistency



Basically Available:

should always be available to respond to user's request even if it can't guarantee immediate access to all data. Can experience small period of data unavailability, but it should be acted upon.

Soft State:

State of database can change over time even without any explicit user intervention. This happens due to background processes, updates to data.

Eventual Consistency:

Database should eventually converge to a consistent state, even if it takes some time to propogate and reflect the changes in data.

eg: Online shopping sites, social media platform

- Use Cases for NoSQL databases

A) Key value database

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1. Session Management on a large scale
2. Using cache to accelerate application response
3. Storing personal data on specific users.
4. Product recommedation
5. Managing each player's session in massive multiplayer online games.

B) Column databases
1. Content management systems
2. Blogging platforms
3. Systems that maintain counters
4. Services that have expired usage
5. Systems that require heavy write requests

C) Document databases
1. Content Management Systems
2. E-commerce product catalogs.

D) Graph database
1. Social Networks
2. Fraud detections

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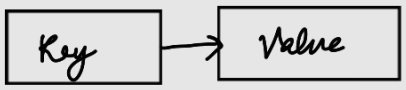
- Limitations of NoSQL databases:
1. Lack of standardization: Diff models and query languages used.
2. Relaxed ACID compliance
3. Complexity in Data Modelling: Developers must design schemas and access patterns based on the application requirements.
4. Tooling and Ecosystem Maturity: Fewer third-party libraries, integrations, community resources available.
5. Consistency trade offs.

- NoSQL architectural Patterns
logical way of categorizing data that will be stored in the database.

1. Key Value store database

Most basic data model which stores data in key-value pairs.

Key: Unique, can be of any data type
Value: Correlated to key (JSON, BLOB etc.)



egs: anna.jpg ⟶ JPG Image file
https://youtube.com ⟶ Webpage

Usually implemented as a hash table where each key is unique.
Used in shopping websites, e-commerce

Advantages:
- Handles large data, heavy load
- Easy retrieval by keys

Disadvantages:
- Low performance
- Complex relationships may arise

2. Column Store Database

Data stored in individual cells which will be handled differently. Individual column will contain several columns inside it. Here, columns are main storage units. They store large volumes of data together.

COLLEGE			
STUDENT			DEPT
Id	Name	Marks	

Advantages:
- Readily available data
- Aggregate queries can run readily on data

Disadvantages:
- Not efficient in online transaction processing.

3. Document database

Stored data in key value pairs but as documents. It can be any data structures like array, string, XML, JSON etc. Can be nested too. It is very effective for unstructured and semi-structured data.

Key	Document
101	{ "Emp_Name": "Anna", "Age": 25 }

Advantages:
- Semi structured data
- Retrieval and management of data is easy.

Disadvantages:
- Aggregate operations may not work fine.

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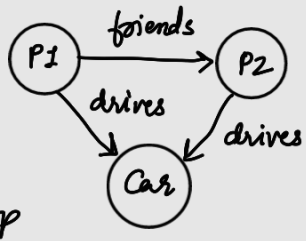
4. Graph database

Storage of data in graphs. Objects as nodes and relationship that define them as edges. A graph depicts connections or relationships b/w objects. Graph databases have strong and direct relationships.

eg: Social N/ws



Advantages:

- Faster traversal, retrieval

Disadvantage:

- Cases of infinite loop

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